INSTRUCTION: Make a Copy, Duplicate this Slide and then Edit!

DeadSwitch — Crypto Asset Inheritance & Digital Will Protocol

VaultVipers

Blockchain and DApps

GDG on Campus

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
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Problem Statement

- What problem are you looking to solve?

An estimated **$140 billion in cryptocurrency is permanently lost** because owners died without a safe way to pass it on.

Three options exist today. None work:

Share your private key while alive → your heir can access funds today, before you're gone. Security collapses entirely.

Don't share anything → you die suddenly, family gets nothing. Mathematically inaccessible forever.

Write keys in a will → wills are public legal documents. Anyone reading probate court records can drain your wallet.

Beyond financial assets — ENS names, NFTs, DAO memberships, DeFi positions — none have any inheritance mechanism. They simply vanish permanently.

Product Idea

- How does your Product solve the problem?

DeadSwitch is a trustless smart contract vault where you deposit crypto, designate heirs with percentage splits, and designate trusted guardians.

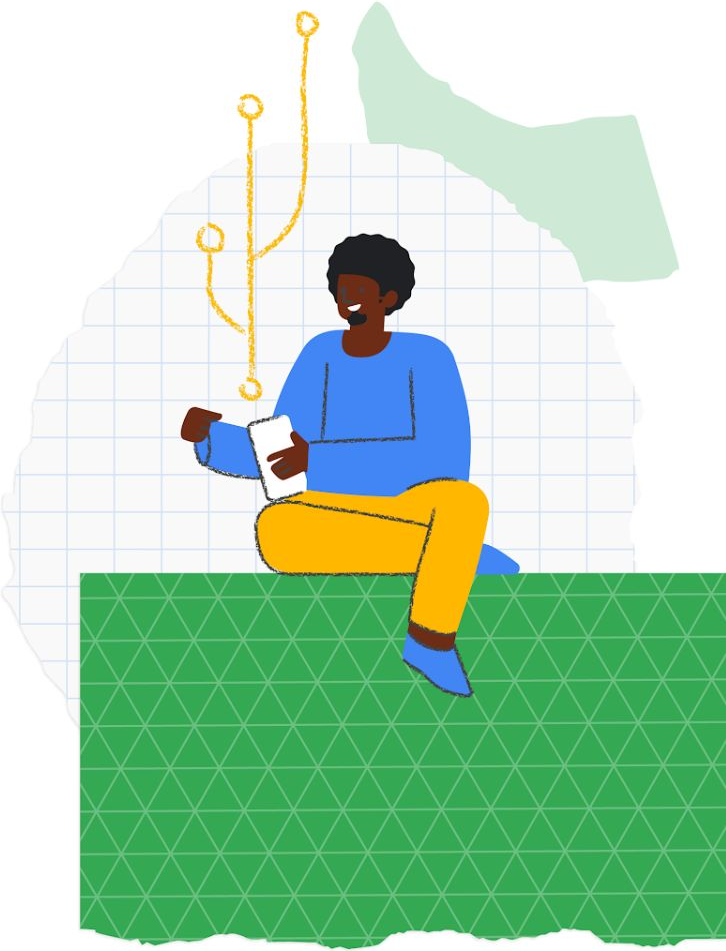
While alive: You check in periodically with one click. Assets stay completely under your control. Nobody else has any authority.

On inactivity: Two-stage trigger. Stage 1 — warning notifications sent to owner only, nothing moves. Stage 2 — after continued silence, guardians are notified and voting begins. Once M-of-N guardians confirm, assets automatically distribute to designated heir wallets exactly as specified.

On recovery: Any guardian can contest instantly, freezing the entire process. A designated emergency contact can pause the vault entirely for medical emergencies — coma, accident, hospitalization.

On completion: Each heir receives a Soul-Bound NFT — a non-transferable, permanently on-chain proof of legitimate inheritance usable with tax authorities and regulated exchanges.

No lawyer. No company. No trust in any third party. The contract is the will.



Additional Info

Dead-Man Switch + Two-Stage Trigger Owner checks in periodically. Stage 1 on inactivity: owner warnings only. Stage 2 after continued silence: guardian voting activates. Medical emergency pause available via single trusted contact.

Guardian Multi-Signature Consensus M-of-N guardians must confirm before a single asset moves. Any guardian contests → process freezes immediately. Guardian reliability flagging alerts owner if a guardian wallet has been completely inactive for 6+ months so they can replace proactively.

ML Behavioral Layer Analyzes owner's on-chain transaction history to detect genuine inactivity. Low risk score → standard owner reminders. High risk score → accelerated escalation to Stage 2. Separately monitors guardian voting timing patterns to detect collusion and immediately notifies owner. Flags unreliable guardians proactively. Emergency pause overrides entire ML pipeline — no ML score delays a legitimate medical pause.

Automated Distribution + Soul-Bound Proof Assets distribute in batches per asset class to designated heirs in exact percentages. Each heir receives a non-transferable Soul-Bound ERC-721 token containing guardian signatures, timestamps, and transfer record — permanent on-chain legal documentation.

How Unique is it?

- Existing Work vs Coined Novelty

Existing Solution	Why It Fails
Share private key	Gives access today, destroys security while alive
Write in a will	Wills are public — anyone can steal keys
Gnosis Safe multi-sig	Co-signers have authority today, no time-trigger, no death condition
Vault12 / Casa	Centralized companies. Can be hacked, shut down, or acquired
Ledger Recover	Shares key shards with third-party companies. Centralized trust

Solves the security-inheritance paradox — sole control while alive, automatic transfer on death. First time this exists trustlessly.

Two-stage trigger — warning period before guardians are ever involved. Prevents false triggers from travel or illness.

Guardian social consensus as death oracle — no centralized service determines death. Trusted humans who know the owner decide. Manipulation-resistant through threshold requirements.

ML collusion detection — statistically flags coordinated guardian voting. Living owners protected from conspiring guardians.

Soul-bound inheritance proof — on-chain legal documentation of legitimate transfer. No existing product generates this.

Multi-asset in one vault — ETH, ERC-20, NFTs with per-asset heir designation simultaneously.

Tech Stack

Blockchain: Solidity 0.8.20, Hardhat, OpenZeppelin (ReentrancyGuard, SafeERC20, ERC721Receiver, Pausable), Polygon Amoy Testnet

Frontend: React 18, Vite, Tailwind CSS, Ethers.js v6, MetaMask, React Router v6, date-fns

Backend: Node.js, Express, Ethers.js v6 (event listener), Telegram Bot API (guardian notifications)

ML: Python, Flask, scikit-learn (behavioral anomaly detection, guardian collusion detection, guardian reliability flagging)

Storage: IPFS, Pinata (vault metadata only — no sensitive data on-chain or IPFS)
Security: Slither static analysis, Hardhat gas reporter, Solidity coverage, custom errors, pull-over-push pattern, batch asset distribution

GDG on Campus

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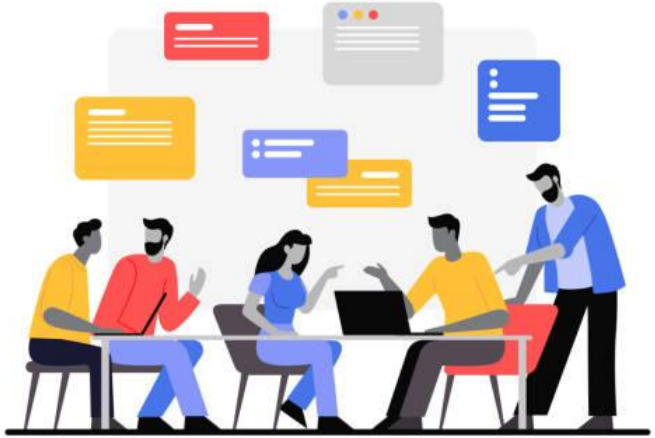
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